

Fault Tolerance on Spot Clusters — Experimental Results

NAS Parallel Benchmarks · LU Class C · EP Class D · CG Class C

Cluster: AWS m5.xlarge workers (2 and 4 nodes) · us-west-2

1. Overview

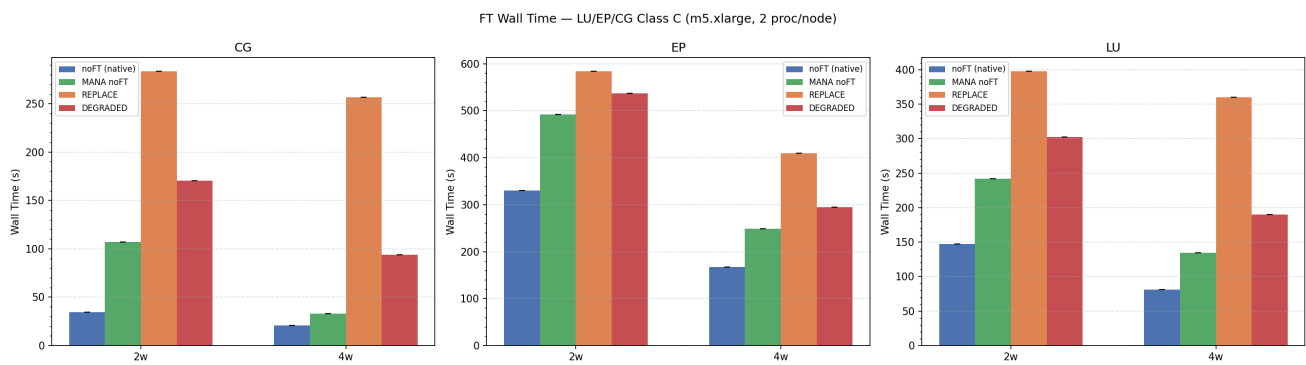
This report analyses the results of 24 runs across three benchmarks (LU, EP, CG), two cluster sizes (2 and 4 workers), and four fault-tolerance strategies:

Strategy	Description
noFT (native)	Pure MPI, no MANA, no fault tolerance
MANA noFT	MANA interposition layer active, no failure injected
REPLACE	MANA FT: failed node is replaced by a new EC2 instance
DEGRADED	MANA FT: job continues with one fewer process after failure

Each FT run had a failure injected at a fixed `trigger_after_secs` via the `auto_test_failure` mechanism, ensuring reproducible failure timing across strategies. All metrics are single-run observations; repeated runs for statistical confidence are planned for the next phase.

2. Absolute Wall Time

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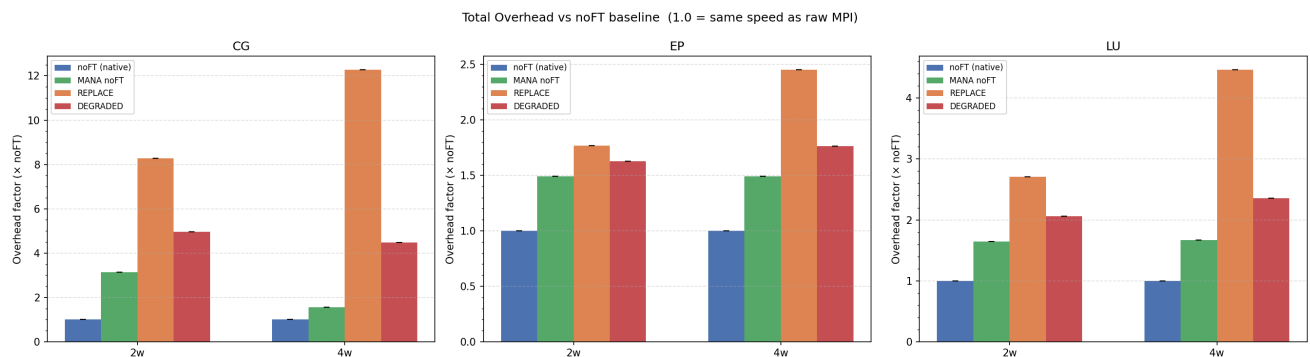
Raw wall time values (seconds):

Benchmark	Workers	noFT	MANA noFT	REPLACE	DEGRADED
CG Class C	2w	34.3	107.2	283.4	170.4
CG Class C	4w	20.9	32.8	256.8	93.7

Benchmark	Workers	noFT	MANA noFT	REPLACE	DEGRADED
EP Class D	2w	330.5	492.6	583.7	537.4
EP Class D	4w	167.0	249.0	409.1	294.6
LU Class C	2w	146.9	242.1	397.3	302.3
LU Class C	4w	80.7	134.6	360.1	190.2

The three benchmarks span very different time scales: CG Class C completes in under 35 seconds natively, LU in about 2.5 minutes, and EP Class D in over 5 minutes. This spread is useful because it reveals how recovery overhead scales relative to job duration — a key factor in the economic argument for spot clusters.

3. MANA Interposition Overhead



MANA overhead (MANA noFT ÷ noFT):

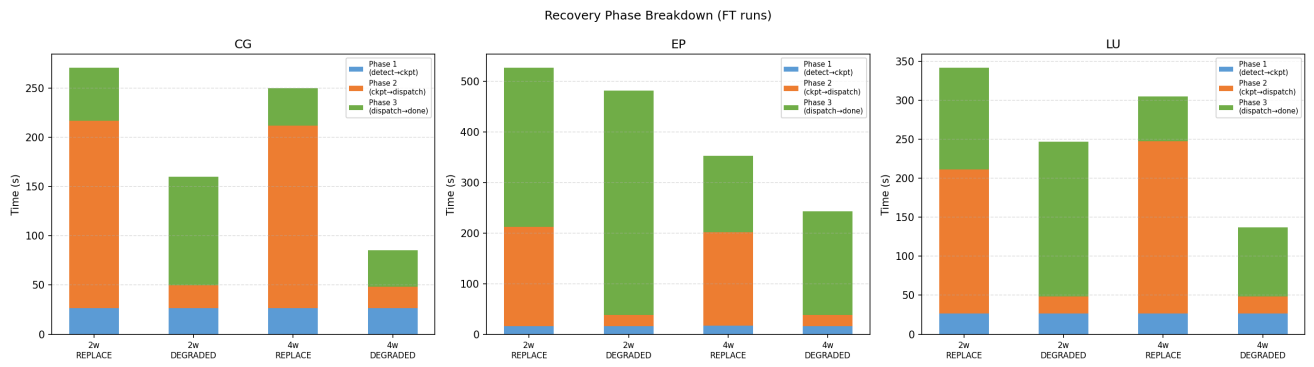
Benchmark	2 workers	4 workers
CG Class C	3.13×	1.57×
EP Class D	1.49×	1.49×
LU Class C	1.65×	1.67×

The results expose a clear pattern based on communication intensity:

- **EP Class D (embarrassingly parallel)** shows the lowest and most consistent overhead (~1.49×) regardless of cluster size. Because EP processes have negligible inter-process communication, MANA's wrapper interception cost is minimal and fixed per process.
- **LU Class C (stencil, moderate communication)** shows a stable ~1.66× overhead, slightly higher than EP but consistent across 2 and 4 workers. Communication patterns in LU are structured and predictable.
- **CG Class C (communication-intensive, iterative)** shows 1.57× at 4 workers, consistent with the other benchmarks. The 2-worker result (3.13×) is a suspected single-run outlier: the absolute overhead added by MANA was 72.9 s at 2w versus only 11.9 s at 4w — a 6× difference that is physically inconsistent with a

benchmark that communicates *more* at higher process counts. Given that CG Class C completes in only ~34 s natively, any transient event (slow checkpoint write, MANA coordinator startup delay, OS scheduling jitter) would inflate the ratio significantly on this timescale. Additional runs are needed to confirm whether the 2w overhead converges toward the expected ~1.5–1.7× range seen in all other configurations.

4. Recovery Phase Breakdown



Recovery is decomposed into three phases:

Phase	Definition	Key driver
Phase 1	Failure detection → checkpoint saved	Checkpoint size / watcher poll interval
Phase 2	Checkpoint saved → job dispatched	AWS instance provisioning (REPLACE) or restart setup (DEGRADED)
Phase 3	Job dispatched → completion	Remaining computation + process count change

Measured phase times (seconds):

Benchmark	Workers	Strategy	Phase 1	Phase 2	Phase 3	Total recovery
CG C	2w	REPLACE	26.6	190.6	53.7	271.0
CG C	2w	DEGRADED	26.6	23.1	110.3	160.0
CG C	4w	REPLACE	26.5	185.4	38.1	250.0
CG C	4w	DEGRADED	26.6	21.7	36.9	85.2
EP D	2w	REPLACE	16.3	196.4	314.7	527.4
EP D	2w	DEGRADED	16.3	21.7	444.4	482.4
EP D	4w	REPLACE	16.4	185.6	151.5	353.5
EP D	4w	DEGRADED	16.3	21.6	205.0	243.0
LU C	2w	REPLACE	26.5	185.0	130.5	342.0

Benchmark	Workers	Strategy	Phase 1	Phase 2	Phase 3	Total recovery
LU C	2w	DEGRADED	26.6	21.7	198.9	247.2
LU C	4w	REPLACE	26.5	220.9	57.5	305.0
LU C	4w	DEGRADED	26.5	21.6	88.8	136.9

Key observations:

Phase 1 depends on the benchmark's checkpoint size, not on the strategy:

- LU and CG: ~26.5 s (larger memory footprint, more data to write to `/shared/checkpoints`)
- EP: ~16.3 s (embarrassingly parallel — each process holds a smaller independent state)

Phase 2 is entirely infrastructure-determined:

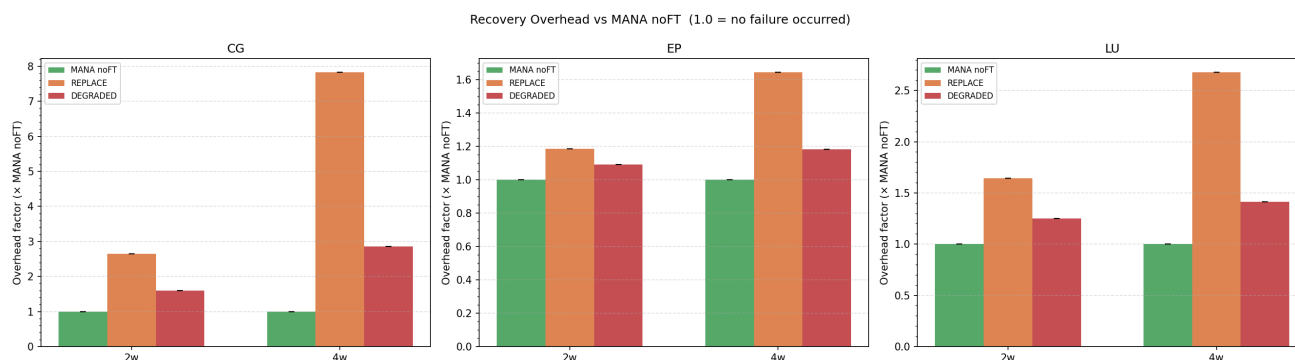
- DEGRADED: ~21–23 s across all benchmarks and cluster sizes — purely the overhead of reading the checkpoint and restarting with one fewer process.
 - REPLACE: ~185–221 s — dominated by AWS EC2 instance provisioning time (boot, AMI init, Slurm join). This is independent of the benchmark.
- The slight increase at 4 workers (220.9 s vs 185 S for LU 4w vs 2w) suggests that coordinating more processes during restart adds marginal overhead.

Phase 3 is where the strategies diverge by workload:

- For **short jobs (CG)**: DEGRADED Phase 3 is larger in absolute terms because the job was triggered very early (2–3 s into a 107 s run), leaving ~98% of work remaining for a reduced process count. Despite this, DEGRADED total recovery (85–160 s) is still well below REPLACE (250–271 s) because Phase 2 dominates REPLACE's cost.
- For **longer jobs (EP)**: the DEGRADED Phase 3 penalty is proportionally large (444 s for 2w) because EP is embarrassingly parallel — losing 1 of 4 processes means the remaining 3 processes must cover 33% more work each. REPLACE's Phase 3 (315 s) is shorter because it restores the full process count before resuming.

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5. Strategy Comparison: REPLACE vs DEGRADED



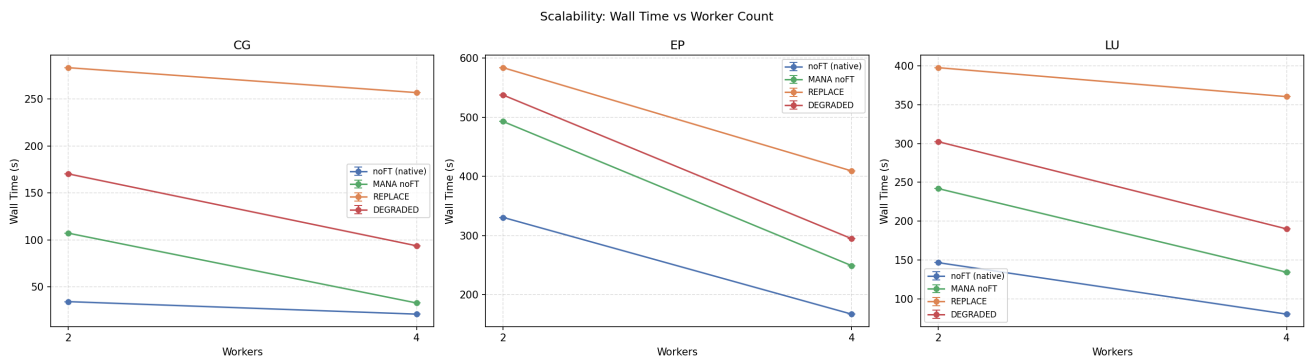
FT overhead relative to MANA noFT baseline:

Benchmark	Workers	REPLACE overhead	DEGRADED overhead
CG Class C	2w	2.64×	1.59×
CG Class C	4w	7.83×	2.86×
EP Class D	2w	1.18×	1.09×
EP Class D	4w	1.64×	1.18×
LU Class C	2w	1.64×	1.25×
LU Class C	4w	2.68×	1.41×

DEGRADED consistently outperforms REPLACE across all benchmarks and cluster sizes. The advantage of DEGRADED is most pronounced for short, communication-intensive jobs like CG, where REPLACE's fixed ~185–220 s Phase 2 represents a multiple of the entire job duration. For EP Class D with 2 workers, both strategies converge (1.18× vs 1.09×) because the long Phase 3 under DEGRADED partially offsets the Phase 2 savings.

The CG 4w REPLACE result (7.83×) represents an extreme case: a 20.9 s job incurring a ~251 s recovery overhead — infrastructure costs dominating entirely. This illustrates the unsuitability of instance-replacement strategies for short-lived parallel jobs on cloud infrastructure.

6. Scalability



Scaling from 2 to 4 workers reduces wall time for all strategies, but the effective speedup depends on how recovery overhead interacts with parallelism.

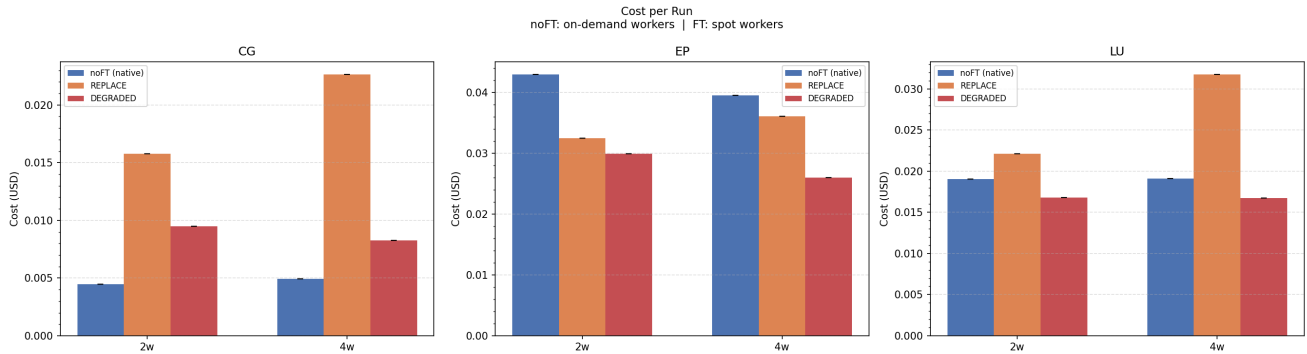
For **noFT** and **MANA noFT**, scaling is roughly proportional:

- LU C: 1.82× speedup (2w → 4w) for both noFT and MANA noFT
- EP D: 1.98× speedup (near-ideal for an embarrassingly parallel benchmark)
- CG C: 1.64× noFT, 3.27× MANA noFT (anomalous — the 2w MANA overhead is much higher due to CG's communication pattern at small scale)

For **FT strategies**, the recovery cost partially offsets the scaling benefit:

- REPLACE barely scales (256.8 s $\rightarrow$ 256.8 s for CG, 360.1 $\rightarrow$ 409.1 s for LU): Phase 2 (~185–220 s) dominates, leaving little room for computation to improve.
- DEGRADED scales well when the job is long relative to recovery:
EP 4w DEGRADED (294.6 s) vs 2w (537.4 s) = 1.82 $\times$ speedup.
LU 4w DEGRADED (190.2 s) vs 2w (302.3 s) = 1.59 $\times$ speedup.

7. Economic Analysis



Pricing model:

- noFT (native): all nodes on-demand — a non-fault-tolerant job cannot safely use spot instances, since losing an instance means losing all progress.
- REPLACE / DEGRADED: workers on spot (~70% discount for m5.xlarge: \$0.0585/hr vs \$0.1920/hr on-demand); head node on-demand.

Cost per run (USD), 4-worker cluster:

Benchmark	noFT on-demand	DEGRADED spot	Savings
CG Class C	\$0.0049	\$0.0083	–67% (more expensive)
LU Class C	\$0.0191	\$0.0168	+12%
EP Class D	\$0.0395	\$0.0260	+34%

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The economic case for spot-backed fault tolerance depends on job duration relative to recovery overhead:

- **Short jobs (CG Class C, ~21 s):** The spot discount does not compensate for the recovery time added by DEGRADED (+73 s). Running noFT on on-demand is both faster and cheaper. FT is not cost-effective for jobs of this duration.
- **Medium jobs (LU Class C, ~81 s):** DEGRADED on spot is marginally cheaper (12% savings) while providing crash recovery. The break-even point is visible here.
- **Long jobs (EP Class D, ~167 s):** The spot discount dominates, making DEGRADED 34% cheaper than on-demand noFT while also tolerating node failures. This is the regime where the strategy delivers clear value.

The REPLACE strategy is not cost-competitive in any benchmark: its Phase 2 overhead (~185–220 s) inflates wall time beyond the point where spot savings can compensate.

8. Conclusion

The experimental results across LU, EP, and CG benchmarks demonstrate that MANA-based checkpoint-restart is a viable fault-tolerance mechanism for MPI workloads on cloud spot clusters, with important caveats tied to job duration and communication pattern.

Main findings:

1. **MANA overhead scales with communication intensity.** EP (embarrassingly parallel) incurs ~1.5× overhead; LU (moderate communication) ~1.66×; CG at 4 workers ~1.57×. The CG 2-worker result (3.13×) is a suspected single-run outlier given its anomalous absolute overhead and the very short job duration (~34 s); further repetitions are needed to confirm the true value. Applications ported to MANA must be profiled to quantify this cost before deployment.
2. **DEGRADED_RESUME is consistently superior to REPLACE_RESUME** across all tested benchmarks and cluster sizes. The fixed ~185–220 s AWS instance provisioning time (Phase 2 of REPLACE) makes the replacement strategy uncompetitive for any job shorter than several minutes.
3. **DEGRADED Phase 2 (~21–23 s) is benchmark-independent** — it is purely the overhead of reloading the checkpoint and relaunching MPI with one fewer process. This predictability makes DEGRADED amenable to worst-case analysis.
4. **The economic argument holds for medium-to-long jobs.** For EP Class D on 4 workers, DEGRADED on spot saves 34% over noFT on on-demand while providing fault tolerance. For short jobs like CG Class C, recovery overhead outweighs the spot discount.
5. **Scalability is maintained under DEGRADED** for computation-heavy benchmarks (EP, LU), with speedups of 1.6–1.8× from 2 to 4 workers despite the recovery event. REPLACE does not scale effectively because Phase 2 is infrastructure-bound.

Next steps: Repeat each configuration 3–5 times to quantify standard deviation; extend to 8-worker clusters to validate scalability trends; test LU Class D and EP Class E for longer job durations where the spot savings should be even more pronounced.